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Can the opening of high-speed rail promote household multidimensional relative poverty alleviation?

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ABSTRACT

Addressing poverty is paramount, aligning with the first Sustainable Development Goal focused on eradicating poverty in all its forms. While the effects of high-speed rail (HSR) on absolute poverty have been documented, its impact on relative poverty remains understudied. This paper examines the influence of HSR on household relative poverty in China through a quasi-experimental design. The main results are as follows: (1) The opening of HSR significantly reduced the household relative poverty by approximately 1.8 %. (2) This alleviation effect primarily transpires through the expansion of economic activities and employment opportunities. (3) Notably, the impact of HSR is more pronounced in lower-ranked, smaller cities and in the western regions of China. Moreover, households with migrant workers or those engaged in non-agricultural sectors derive greater benefits from HSR developments. Our results suggest that HSR opening could have contributed to China's relative poverty alleviation. Policymakers can consider the role of transportation infrastructure in mitigating household relative poverty, especially for low rank cities, small cities and periphery regions in other developing countries.

1. Introduction

The first Sustainable Development Goal (SDG) is tasked with eradicating poverty in all its dimensions (Pomati and Nandy, 2020). The most widely used measure of global poverty involves the proportion of people living below the World Bank's $1.90 a day line (previously $1.25 in 2005 ICP prices), representing both an absolute and a minimal threshold. With absolute poverty addressed in some countries, policymakers are now increasingly focusing on relative poverty (Foster, 1998; Wang and Feng, 2020).

Relative poverty is conceptualized as an individual's inability to meet essential life needs compared to societal averages, spotlighting issues of inequitable social distribution (Wang and Alkire, 2009). This form of poverty extends beyond mere economic deprivation, indicating a failure to integrate disadvantaged populations into the societal mainstream, thereby marginalizing them in aspects of production, livelihood, and social engagement (Zhang et al., 2024). The persistence of relative poverty, even in contexts where absolute poverty has been eradicated, underscores its roots in uneven societal development (Zhang and Shen, 2020). Addressing relative poverty not only directly contributes to diminishing global poverty but also facilitates the achievement of related Sustainable Development Goals. To a certain extent, relative poverty can be attributed to some external factors (Fan and Bai, 2023; Meng et al., 2023; Xie et al., 2023; Zhou and Shao, 2020; Zong et al., 2023), and transportation infrastructure is one of them.

Transportation infrastructure, especially high-speed rail (HSR), plays a pivotal role in enhancing residents' quality of life through improved access to jobs, services, and increased income (Di Matteo and Cardinale, 2023; Kim, 2000; Lu et al., 2022b, 2023; Shaw et al., 2014; Shi et al., 2020). The strategic deployment of HSR over the past decades has not only reshaped transportation networks but also spurred socioeconomic advancements and income enhancements (Chen et al., 2021; Shi et al., 2020; Sun and Mansury, 2016; Wang et al., 2019; Zhang et al., 2022). Despite extensive research validating HSR's role in boosting regional connectivity and economic development, its impact on alleviating relative poverty remains less explored, particularly in contrast to its effects on absolute poverty as demonstrated through various economic indicators (Heuermann and Schmieder, 2019; Kaewunruen et al., 2019; Li et al., 2020; Liang et al., 2020; Wu et al., 2022; Zhang et al., 2023). Existing research related to relative poverty only investigated the reduction effect from economic factors, such as labor migration (Fan and Bai, 2023; Zong et al., 2023), non-agricultural employment (Zhou and

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Shao, 2020), digital economy (Meng et al., 2023; Wei and Wei, 2023), and digital finance (Peng and Mao, 2023; Xie et al., 2023). However, the relationship between HSR opening and resident relative poverty is still unclear.

This research selects China as our case for several reasons. First, China is the biggest developing country in the world, which has prioritized poverty reduction in recent decades, transitioning from eradicating absolute poverty to addressing relative poverty post-2020 (Wang and Sun, 2021; Wang and Feng, 2020). Second, as a vital strategic asset for poverty alleviation in China, HSR has seen rapid expansion over the past decades. By the end of 2022, the operational mileage of China's HSR network reached 42,000 km, the most extensive globally, and has progressively extended from major cities to county-level areas in peripheral regions (Xinhua News Agency, 2023), thus playing a crucial long-term role in facilitating poverty reduction (Zhang et al., 2023).

To examine whether the rapid development of HSR in China has a positive effect on alleviating relative poverty among residents, this study used household data from the China Family Panel Studies (CFPS). We also construct the multidimensional relative poverty (MRP) indicator system to measure the relative poverty level of residents' households based on the A-F method (Alkire et al., 2022; Alkire and Foster, 2011). We further investigate the impact of HSR opening on household relative poverty, and examine the mechanisms and impact heterogeneity.

The main contributions of this study are as follows. (1) It evaluates the impact of HSR services on household multidimensional relative poverty, enriching the existing literature on the nexus between transportation infrastructure and relative poverty. (2) It adopts a multidimensional approach to poverty measurement using micro-level data, thereby addressing the limitations of conventional poverty indicators such as income levels and the Gini coefficient, which often result in biased poverty assessments and overestimated effects of HSR on poverty reduction. (3) This study further examines the mechanisms of HSR in reducing household relative poverty, we find that HSR particularly contributes to promoting local economic development and labor employment.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on transportation infrastructure and poverty reduction. Section 3 reviews the background of China's HSR development and poverty governance. Section 4 explains data sources, variable specification and model setting. Section 5 presents the empirical results and conducts robustness tests. Section 6 discusses the results, synthesizes the key findings and provides policy implications.

2. Literature review

The existing literature extensively discusses the impact of transportation infrastructure development. Previous research has explored the influence of transportation infrastructure at the macro level, such as spatial structure (X. Gao et al., 2022b; Huang and Zong, 2020), economic development (Campos and De Rus, 2009; Yang et al., 2022) and industrial structure (Jin and Zhu, 2023; Zhou and Zhang, 2021). Some studies also have analyzed the effects of HSR at the micro-level, such as individual mobility, labor employment, resident income, etc. (Di Matteo and Cardinale, 2023; Kim, 2000; Shi et al., 2020; Willigers and van Wee, 2011; Zhang et al., 2022). However, only a few scholars have explored the impact of HSR on poverty (Li et al., 2020; Wu et al., 2022; Zhang et al., 2023), and there is still a lack of research related to its impact on relative poverty.

2.1. Impact of transportation infrastructure on poverty reduction

In the literature concerning transportation infrastructure and its role in poverty reduction, some scholars posit that enhancements in transportation infrastructure significantly contribute to alleviating poverty (Cai et al., 2023; Yang and Song, 2022). They argue that enhancements in transportation accessibility facilitate the breakdown of geographical barriers, improve regional connectivity, and promote equitable economic development (Cascetta et al., 2020; Huang and Zong, 2021; Yang et al., 2018). Specifically, transportation infrastructure can facilitate the labor mobility between urban and rural areas. This transition aligns with Lewis (1954), which posits that labor moves from sectors with excess labor and lower wages to those with higher wages and productivity, thus raising overall income levels. Furthermore, infrastructure developments, such as roads and railways, reduce transportation costs, increase access to markets, and consequently enhance income opportunities for both urban and rural populations (Lu et al., 2022b, 2023; Shaw et al., 2014; Wondemu et al., 2012). Specific studies highlight that road infrastructure investments significantly reduce absolute poverty by enhancing access to essential services and improving agricultural productivity and technology (Aggarwal, 2018; Fan and Chan-Kang, 2008; Jiang et al., 2020; Sewell et al., 2019; Warr, 2008). For instance, Fan and Chan-Kang (2008) suggested that the poverty-alleviating effect of road opening particularly benefits the less economically-developed regions. Similarly, Tian et al. (2024) found that highway opening has significantly reduced the multidimensional poverty at the county level in China in recent years.

In relation to HSR, evidence suggests that it contributes to poverty reduction by enhancing regional accessibility, which facilitates job creation, boosts employment rates, and increases income for impoverished households (Li et al., 2020; Lu et al., 2022a; Zhang et al., 2023). On the one hand, HSR facilitates higher wage earnings for households by creating jobs and enabling the transfer of employment to more productive sectors, particularly benefiting non-agricultural sectors (Jiao et al., 2025; Li et al., 2020; Qin, 2016; Wu and Liu, 2022; Zhang et al., 2023). On the other hand, HSR is instrumental in enhancing agricultural incomes, a critical factor in poverty alleviation (Montalvo and Ravallion, 2010; Qin, 2016). The deployment of HSR supports the dissemination of agricultural innovations and the migration of skilled agricultural workers, thereby boosting agricultural productivity and facilitating the transition of surplus rural labor toward more productive sectors (Liu and Guo, 2019; Lu et al., 2022b; Wang et al., 2020). These effects are particularly significant in promoting poverty alleviation in marginal and impoverished regions (Lu and Li, 2025). Nonetheless, the specific effects of HSR on relative poverty remain an area for further investigation.

2.2. Conceptualization and measurement of relative poverty

Poverty measurement is based on a comparision of resources to needs (Foster, 1998). Relative poverty is characterized by an individual's inability to access the resources necessary to maintain a standard of living comparable to that of their societal peers (Wang and Alkire, 2009). The original concept of relative poverty extends beyond mere financial insufficiency and incorporates the cost of social inclusion, which includes the ability to participate in key societal activities such as the labor market (Sen, 1983). Later, it further included other multidimensional poverty dimensions, such as social health, living conditions, education, etc. (Alkire et al., 2022; Zou et al., 2023).

Relative poverty offers several advantages over absolute poverty measurements. First, within a multidimensional framework, it emphasizes social exclusion and inequality within specific contexts. It highlights whether individuals or households are deprived relative to others in their society (Foster, 1998; Wang and Alkire, 2009). This is crucial because individuals with a basic income may still experience poverty if they lack access to essential services or live in an environment marked by high inequality (Garroway and de Laiglesia, 2012). Second, while absolute poverty is often a static measure with fixed thresholds (Foster, 1998; Wang and Alkire, 2009), relative poverty is more dynamic. It adapts to changes in general living standards, making it more sensitive to evolving social needs and their associated costs. Income-based indicators are static and capture only economic capacity, while the Gini coefficient reflects solely income inequality. These measures represent just one dimension of human development and poverty, failing to account for deprivations and disparities in access beyond income (Fisher, 1997). As a result, they are inherently limited in identifying groups suffering from broad but pervasive forms of deprivation (Alkire and Foster, 2011). Relative poverty lines, therefore, better reflect shifts in societal conditions across countries and over time, as they change in response to broader social and economic transformations (Garroway and de Laiglesia, 2012).

Currently, various methodologies can be employed to measure relative poverty, including the income ratio method and the multidimensional poverty index. The income ratio approach, which principally assesses economic disparities, identifies those earning below 40 % to 60 % of the median or average income as experiencing poverty (Chen et al., 2013; Madden, 2000; Sun and Xia, 2019). However, its application has been limited due to its singular focus on economic factors (Ravallion and Chen, 2019; Zhang and Shen, 2020).

The multidimensional approach, including indices like the Hagenaars-Maître (H-M) index, the Human Poverty Index (HPI), and notably the Alkire-Foster (A-F) method, provides a more comprehensive poverty assessment (Ding, 2014). Proposed by Alkire and Foster in 2007, the A-F method has been adopted by many countries to measure the multidimensional poverty level of residents (Alkire and Foster, 2011; Wang and Sun, 2021; Wang and Feng, 2020). This method mainly includes the objective and subjective dimensions. Specially, objective dimensions include indicators related to health, income, education, living standards, etc. (Dong et al., 2021; Zou et al., 2023). Subjective indicators are measured based on resident's perceptions of their living situation and well-being, such as income adequacy and life satisfaction (Y. Gao et al., 2022a). Moreover, this method emphasizes the sustainability and stability of poverty assessments, which helps in avoiding the overestimation of poverty reduction effects (Chen et al., 2013; Dong et al., 2021).

The opening of HSR can impact household relative poverty in the following ways. Generally, it influences the spatial distribution of production factors—such as investment, technology, and knowledge—and shapes industrial development patterns (Heuermann and Schmieder, 2019; Jin and Zhu, 2023; Liang et al., 2020; Zhou and Zhang, 2021). By driving local investment and redistributing industrial growth, HSR promotes new businesses and industrial upgrading, generating substantial local employment opportunities that directly benefit the residents (Diao, 2018; Dong, 2018; Lan et al., 2023; Liang et al., 2020; Yi et al., 2025; Zhang and Xu, 2024). HSR also significantly reduces labor mobility costs and expands labor market boundaries (Fan et al., 2012; Heuermann and Schmieder, 2019; W. Sun et al., 2022b). This facilitates spatial redistribution of the population, enabling labor to seek better job opportunities and improved living resources across regions (Deng et al., 2019; Zhang and Xu, 2023, 2024). Notably, HSR addresses barriers faced by socially vulnerable groups, whose social participation is often limited by lack of opportunities and access. Improved transportation accessibility brought about by HSR can substantially enhance their ability to engage in societal functions and access essential resources (Baş and Delaplace, 2023; Chen et al., 2021). Thus, HSR can alleviate household relative poverty, although the impact may vary heterogeneously.

3. Background

3.1. HSR development in China

China's HSR network represents one of the most rapidly constructed transportation infrastructures in modern history (Chen and Haynes, 2017) (See Fig. 1). From the 1990s, China has developed the world's largest HSR system (Qin, 2016). In 2020, the network connected all Chinese cities with populations over one million, covering 90 % of the national population (Liang et al., 2020). By 2022, operating mileage increased to 42,000 km, with China ranking first worldwide in total railway length, passenger turnover, and transport density (Xinhua News Agency, 2023). According to the 2021 National Comprehensive Transportation Network Plan Outline, HSR mileage is projected to reach 70,000 km by 2035 (CPCCC and SC, 2021). China's nearly four decades of infrastructure investment have significantly influenced economic activity distribution, poverty reduction, and economic integration (Qin, 2016). However, since the HSR system prioritizes densely populated regions, network density and growth have been faster in northern, eastern, and southern China than in the less populated western areas (Chen and Haynes, 2017).

3.2. Poverty governance in China

China's poverty reduction efforts over the past few decades are considered one of the most remarkable social and economic achievements in China (Tian et al., 2024; Zhang et al., 2023). Since the opening up policy in the late 1970s, the living conditions of residents have been continuously improved. China has transformed from one of the poorest nations into the world's second-largest economy. A major milestone was achieved in 2020, China announced that the complete eradication of absolute poverty, lifting 98.99 million rural residents, 128,000 poor villages, and 832 state-designated impoverished counties out of poverty (see Fig. 2). With the elimination of absolute poverty, the focus of China's poverty policy has shifted toward addressing relative poverty after 2020 (Wang and Sun, 2021).

4. Data and methods

4.1. Data

The primary dataset for this study is derived from the CFPS, which began in 2010 and conducts biennial surveys to gather data on households and individuals across 21 provinces and four direct-administered municipalities (Beijing, Shanghai, Tianjin, and Chongqing). This dataset has been extensively utilized in studies examining relative poverty in China (Li et al., 2023; Peng, 2022; Wang et al., 2022, 2021). Our analysis specifically includes households with individuals aged 16 and older who consistently participated in the surveys from 2012 to 2020. Due to the absence of certain critical variables in the 2010 survey, our sample consists of 1902 households surveyed from 2011 to 2019.[1] Additionally, municipal-level socio-economic data was sourced from the "China Urban Statistical Yearbook" covering the same period.

The HSR data are collected manually, and the key steps are as follows: Firstly, information about HSR lines was extracted based on the "Eight Vertical and Eight Horizontal" network strategy outlined in the "Medium and Long-term Railway Network Plan" of 2016. Secondly, details concerning HSR stations, specifically those servicing trains prefixed with "G", including station names, line names, and opening dates, were subsequently gathered. Finally, This geographical and operational data was then mapped and analyzed using ArcGIS to associate each HSR station with its respective county, city, and province.

4.2. Variables

4.2.1. Dependent and independent variables

The dependent variable in our study is the household multidimensional relative poverty index, calculated using the A-F method's multidimensional relative poverty indicator system (Peng, 2022; Wang et al., 2021) (See Table 1). The household multidimensional relative poverty quantifies deprivation across various dimensions, with scores ranging

[1] For the CFPS data used, 2012, 2014, 2016, 2018, and 2020 are the survey years for this dataset. The actual report is the situation of the previous year in the survey year, so we finally obtained micro data for five periods from 2011 to 2019.

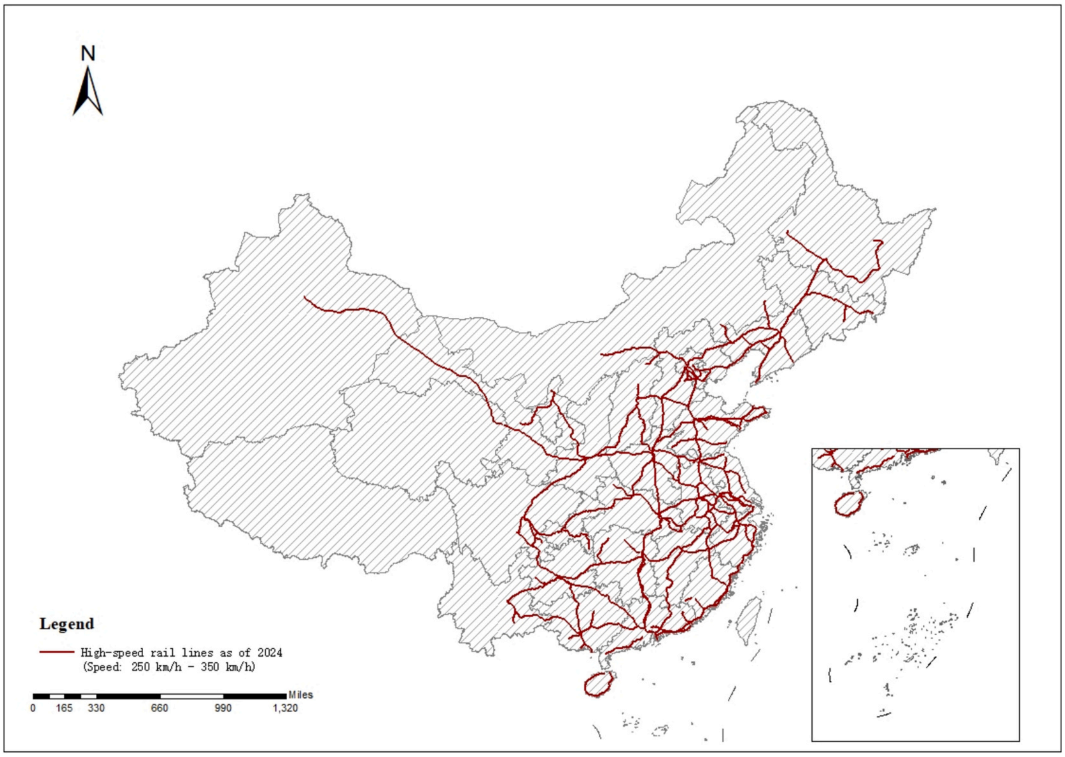

Fig. 1. Distribution map of China's HSR lines as of 2024.

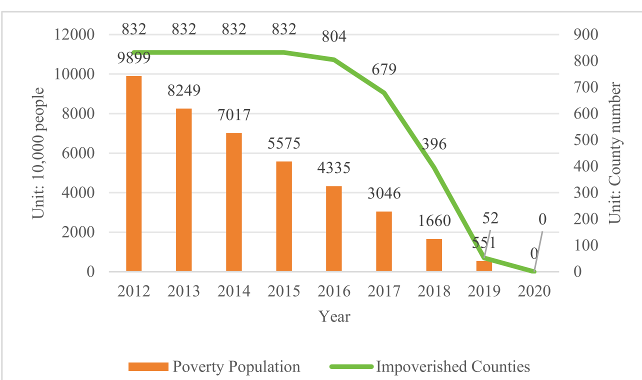

Fig. 2. China's rural poverty population and impoverished counties from 2012 to 2020.

from 0 to 1, where a higher score indicates greater poverty (Dong et al., 2021; Guo et al., 2022; Huang et al., 2023; Liu and Wang, 2019). The calculation of multidimensional relative poverty involves setting deprivation cutoffs and weights for each indicator within the dimensions of health, economy, living conditions, and subjective well-being, following methodologies established by previous research (Alkire et al., 2022; Cheng et al., 2021; Huang et al., 2023; Zhu et al., 2021).

Specifically, in this study, the household multidimensional relative poverty is composed of four dimensions (economic, health, life, and subjective) and ten indicators (see Table 1). The indicators use a nested weight structure based on the A-F dual-cutoff identification approach: equal weights across dimensions and an equal weight for each indicator within a dimension (Alkire et al., 2022; Alkire and Foster, 2011). For instance, the health dimension accounts for 1/4 of the total weight, and, since it includes two indicators, each health-related indicator is assigned a weight of $1/(4 \times 2) = 1/8$. This equal weight can reduce subjective intervention in constructing multidimensional indicator systems (Decancq and Lugo, 2013). The household multidimensional relative poverty is the final deprivation score for each household, which is obtained by summing the weights of all indicators in which the household is deprived (Alkire et al., 2022). As shown in Table 2, the multidimensional relative poverty scores range from 0 to 0.92, indicating substantial variation in the degree of relative poverty across households.

Independent variable is HSR opening, coded as a dummy variable where 1 indicates the opening of an HSR station in the city within the observation period and 0 otherwise. To account for the time-lagged effects of transportation infrastructure on economic outcomes, the opening date is adjusted by one year.[2] The mean value of this variable is 0.48, suggesting that nearly half of the sampled households reside in cities with HSR access during the study period.

4.2.2. Control variables

To minimize potential biases from omitted variables and other extraneous factors, several control variables are included at both the household and municipal levels. At the household level, family size is controlled as it influences poverty risk (Bian and Ji, 2021; Guo et al., 2022; Qian and Zhang, 2023; Wang et al., 2022). Average age of household adults is also included as it influences household income potential (H. Sun et al., 2022a; Wang et al., 2022). The educational attainment within the household, measured by the presence of college-educated members, is also considered, reflecting its impact on employment and income opportunities (H. Sun et al., 2022a; Sun and Li, 2022). Additionally, household financial, social and natural capitals are controlled to capture the resources owned by households. Specifically, financial capital is measured by the total value of household cash and deposits (Bian and Ji, 2021; Sun and Li, 2022). The social capital is

[2] Additionally, as CFPS collect data every two years, we classify HSR opening in even year into the subsequent odd year to match with CFPS data. For example, if a city opened the HSR with the prefix "G" in 2012, the HSR variable will take a value of 1 in 2013 and subsequent years, otherwise, it will take a value of 0 in 2013.

Table 1
The multidimensional index system for measuring household multidimensional relative poverty.

Dimension	Indicator	Indicator Deprivation Threshold	Metric	Weight	Ref.
Economic dimension	Proportion of entertainment expenditure	If the ratio of household cultural, educational and entertainment expenditures to total household expenditures is less than 5 %, it is assigned a value of 1, otherwise 0.	Absolute	1/12	Zhao and Xu (2021)
	Net income per capita	If the per capita net income of the family is less than 40 % of the full sample median in that year, it is assigned a value of 1, otherwise 0.	Relative	1/12	Cheng et al. (2021)
	Mortgage loan ratio	If the household mortgage loan expenditure accounts for more than 40 % of the household net income, it is assigned a value of 1, otherwise 0.	Absolute	1/12	Fan and Bai (2023)
Health dimension	Health status	If the proportion of self-rated unhealthy adults in the family is higher than the mean value of the entire sample of families in that year, it is assigned a value of 1, otherwise 0.	Relative	1/8	Cheng et al. (2021); Fan and Bai (2023)
	Medical insurance	If the coverage rate of adult medical insurance in the family is lower than the mean value of the entire sample of families in that year, it is assigned a value of 1, otherwise 0.	Relative	1/8	Cheng et al. (2021)
Life dimension	Cooking water	If the main water used for household cooking does not include tap water, mineral water, bottled water, purified water, or filtered water, it is assigned a value of 1, otherwise 0.	Absolute	1/8	Cheng et al. (2021); Huang et al. (2023); Zou et al. (2023)
	Cooking fuel	If the main fuel for household cooking does not include electricity, gas, liquefied gas, natural gas, solar energy, or biogas, it is assigned a value of 1, otherwise 0.	Absolute	1/8	Cheng et al. (2021); Huang et al. (2023); Zou et al. (2023)
Subjective dimension[a]	Life satisfaction	If the average self-evaluation score of life satisfaction among household members is below 3, assign a value of 1 (score range of 1–2), otherwise 0 (score range of 3–5).	Absolute	1/12	Fan and Bai (2023); Zhao and Xu (2021)
	Income status	If the average self-assessment score of household members for their own income in the local area is below 3, assign a value of 1 (score range of 1–2), otherwise 0 (score range of 3–5).	Absolute	1/12	Zhao and Xu (2021)
	Social status	If the average self-assessment score of household members for their social status in the local area is below 3, assign a value of 1 (score range of 1–2), otherwise 0 (score range of 3–5).	Absolute	1/12	Zhao and Xu (2021)

[a] The scores for the corresponding questions in the CFPS questionnaire are represented by 1,2,3,4,5, with a score of 3 representing a moderate level. For example, the corresponding question of life satisfaction is: "How do you rate your satisfaction with your life?" Option: 1–2–3-3-4-5. Among them, 1 represents very dissatisfied, 5 represents very satisfied.

Table 2
Summary statistics of key variables.

Variable Type	Variable	Variable Full Name	Unit	Obs	Mean Value	Std. Dev.	Minimum Value	Maximum Value
Dependent variable	*MRP*	Multidimensional relative poverty deprivation score	–	9510	0.335	0.167	0.000	0.917
Key independent variable	*HSR*	The opening of High-speed rail	–	9510	0.483	0.500	0.000	1.000
Control variables	*lnSize*	ln (household size)	Pcs	9510	1.354	0.435	0.000	2.708
	lnAge	ln (average age of household)	Year	9510	3.802	0.212	2.862	4.500
	Fhcl	Household education level	–	9510	0.220	0.414	0.000	1.000
	lnFin	ln (household financial capital)	Yuan	9510	7.319	4.564	0.000	14.914
	lnSoc	ln (household social capital)	Yuan	9510	7.910	1.131	0.000	11.310
	lnLand	ln (household land capital)	Yuan	9510	6.607	4.851	0.000	14.955
	lnTech	ln (technology investment level)	%	9510	−4.624	0.839	–	–
	lnFgn	ln (foreign investment level)	%	9510	0.038	0.044	–	–
	lnExp	ln (public finance expenditure)	Ten Thousand Yuan	9510	14.597	0.639	–	–

Due to restrictions on the use of confidential CFPS data, the maximum and minimum values of some variables cannot be reported.

represented by annual household transportation and communication expenditures (Bian and Ji, 2021). The natural capital is measured by household land assets (Jia et al., 2021), which is an important resource endowment for households relying on agricultural production (Barbier, 2019).

At the municipal level, technology investment, foreign investment and government financial expenditure are chosen for the following reasons. First, the technology investment level indicates a city's innovation capacity, which can significantly affect local economic conditions and income levels (Hu et al., 2022; Yu and Pan, 2019). Second, the degree of foreign investment reflects a city's economic openness and its ability to generate employment, thereby influencing economic development and poverty reduction strategies (Wei and Wei, 2023; Zhang et al., 2023). Third, government financial expenditure is considered due to its role in shaping policies and initiatives aimed at reducing relative poverty within municipalities (H. Sun et al., 2022a; Zhang et al., 2023).

4.3. Model specification

Traditional Difference-in-Differences (DID) methods assume that all treated units are exposed to the policy shock at the same time. The core idea of the multi-period DID model is to treat the timing of policy adoption for each treated group as its individual treatment period, estimate group-specific treatment effects, and then aggregate them to obtain the overall effect.

In this study, the opening of HSR is treated as a quasi-experimental design, where cities with newly opened HSR lines are categorized as the treatment group and those without HSR form the control group. Due to the staggered introduction of HSR across different cities, we employed a multi-period DID approach to estimate the impact of HSR on household relative poverty. The multi-period DID model accounts for both household and year fixed effects, and is specified as follows:

$$MRP_{i,j,t} = \alpha + \beta HSR_{i,t-1} + \theta X_{i,t} + \delta Y_{j,t} + \mu_j + \lambda_t + \varepsilon_{i,j,t} \quad (1)$$

where $MRP_{i,j,t}$ denotes the level of relative poverty of household j in city i at year t; the dummy variable $HSR_{i,t-1}$ indicates whether city i opened the HSR in period $t-1$ or not; $X_{i,t}$ and $Y_{j,t}$ represent control variables; μ_j denotes household fixed effects, λ_t represents year fixed effects, $\varepsilon_{i,j,t}$ represents the error term.

5. Results

5.1. Baseline results

This section presents the empirical results of the multi-period DID regressions. In Table 3, the coefficient of *HSR* is significantly negative at the 1 % level, which indicates the opening of HSR is effective in reducing household relative poverty. Furthermore, the coefficients of *lnFin* and *lnSoc* are significantly negative, suggesting that enhancements in these domains contribute to reducing relative poverty level.

To more intuitively assess the impact of HSR on household multi-dimensional well-being, we examine four variables: wage income (*lnwage*) (Wang et al., 2024), the proportion of unhealthy members (*lnpoor_health*) (Yang et al., 2025), the value of durable goods (*lndurables_asset*) (Zhang et al., 2025), and subjective life satisfaction (*lnswb*) (Chen and Chen, 2023). As shown in Table 4, column (1) reveals that HSR promotes wage income growth, indicating HSR can enhance household economic status. Column (2) indicates that HSR significantly reduces the share of unhealthy household members at the 1 % level, suggesting it improves household health status. Column (3) shows HSR is associated with a significant increase in household durable assets, which reflects HSR can improve household living conditions. Column (4) reports a positive effect of HSR on subjective life satisfaction, although the coefficient is insignificant. These results support that HSR contributes to multidimensional poverty alleviation by improving household economic status, health, living standards, and subjective well-being. Such effects are likely to be driven by improved access for residents to socioeconomic resources, health facilities due to HSR opening (Alhassan and Anciaes, 2025; Wang et al., 2025; Yang et al., 2025).

Table 3
Baseline regression results.

	(1)	(2)
Dependent variable:	*MRP*	*MRP*
HSR	−0.016***	−0.018***
	(−2.62)	(−3.00)
lnSize		0.021**
		(2.33)
lnAge		0.113***
		(7.00)
Fhcl		−0.005
		(−0.74)
lnFin		−0.002***
		(−4.90)
lnSoc		−0.008***
		(−3.98)
lnLand		0.000
		(0.17)
lnTech		0.009*
		(1.92)
lnFgn		−0.303
		(−1.48)
lnExp		−0.021
		(−1.13)
Constant	0.357***	0.351
	(105.62)	(1.20)
Observations	9510	9510
R square	0.084	0.107
Number of households	1902	1902
Household fixed effects	YES	YES
Year fixed effects	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

Table 4
Sub-indicator regression results.

	(1)	(2)	(3)	(4)
Dependent variable:	*lnwage*	*lnpoor_health*	*lndurables_asset*	*lnswb*
HSR	0.276*	−0.021***	0.169**	0.003
	(1.88)	(−2.65)	(1.98)	(0.43)
Control variables	YES	YES	YES	YES
Observations	9509	9510	9506	9510
R square	0.096	0.037	0.043	0.210
Number of households	1902	1902	1902	1902
Household fixed effects	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

5.2. Parallel trends test

The key assumption underlying the specification of multi-period DID model is that the stability of relative poverty trends between the treatment (cities with HSR) and control groups (cities without HSR) in the absence of the intervention. The assumption requires that, in the absence of treatment, the average outcome for the treatment and control groups would have followed parallel paths over time. To validate this assumption, the parallel trend test is conducted by adopting the earliest period before the opening of HSR as the base period. Specifically, we replace the dummy variable $HSR_{i,t-1}$ with six dummy variables (including *pre2, pre1, current, post1, post2, post3+*). These dummy variables cover the event window from four years before the HSR opening to six years after HSR opening. The six years and beyond before the opening of HSR serve as the benchmark period. If the dummy variables prior to the treatment are not significant but becomes significant after HSR opening, it suggests that the parallel trends assumption holds.

In Table 5, the coefficient for *HSR* is statistically insignificant in the years leading up to the opening of the HSR—specifically, four and two years prior (also see Fig. 3). This lack of significance suggests that prior to the introduction of HSR, there were no significant differences in poverty trends between the groups, thus supporting the validity of the DID model. However, in the year HSR was opened, the coefficient becomes significantly negative, and this effect persists, being observable in

Table 5
Parallel trends test results.

	(1)
Dependent variable:	*MRP*
The four years before the opening of HSR (*pre2*)	0.001
	(0.12)
The two years before the opening of HSR (*pre1*)	−0.018
	(−1.55)
The year of the opening of HSR (*current*)	−0.035***
	(−2.64)
The two years after the opening of HSR (*post1*)	−0.033**
	(−2.06)
The four years after the opening of HSR (*post2*)	−0.049***
	(−2.74)
The six years and beyond after the opening of HSR (*post3+*)	−0.053***
	(−2.61)
Control variables	YES
Observations	9510
R square	0.109
Number of households	1902
Household fixed effects	YES
Year fixed effects	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

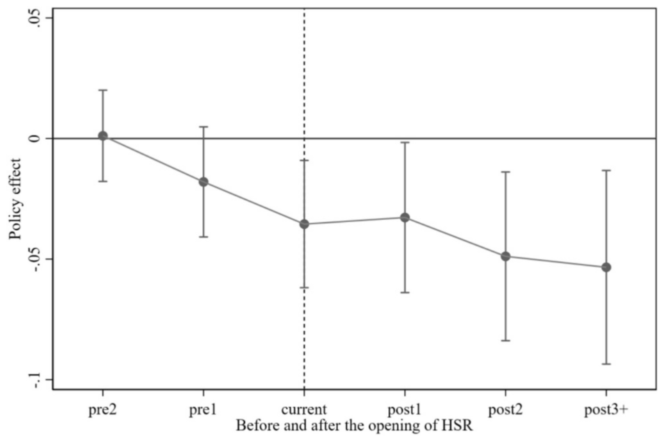

Fig. 3. Parallel trends test results.

two, four, and six years following HSR implementation. The results imply that the opening of HSR has a lasting impact on reducing household relative poverty.

5.3. Mechanism analysis

The impact of HSR opening on the household multidimensional relative poverty can be traced through at least two primary mechanisms. First, HSR construction helps stimulate economic vitality by promoting new businesses and industrial upgrading, thereby creating more employment opportunities. HSR supports the growth of new businesses and drives industrial upgrading in HSR-connected cities (Lan et al., 2023; Liang et al., 2020; Liu and Lin, 2025; Yi et al., 2025). Specifically, HSR can facilitate the flow of capital and technology, enhance knowledge spillovers (Sun et al., 2023; Yang et al., 2022) and promote the spatial reallocation of industries (Chang et al., 2021; Shao et al., 2017; Shi and Wang, 2024). This creates new jobs and raises non-agricultural employment shares (Dong, 2018; Liang et al., 2020; Zhang and Xu, 2023, 2024). Second, HSR development broadens the labor market by facilitating local employment and labor mobility. By significantly shortening the spatial and temporal distance, HSR expands the geographic scope of employment opportunities available to the labor force. This expansion is conducive to meeting the diverse employment needs of individuals, ultimately leading to increased employment opportunities and higher wage income (Di Matteo and Cardinale, 2023; Heuermann and Schmieder, 2019; Kaewunruen et al., 2019). To analyze this effect, the study specifically focuses on the impact of HSR openings on local employment and labor transfer.

In Table 6, the variable *lnnew_ent* represents the number of newly registered enterprises in a city, measured by the total number of newly registered firms across all industries. The results in column (1) indicate that the opening of HSR significantly promotes the growth of new enterprises, suggesting that HSR investment contributes to stimulating local entrepreneurial vitality and economic activities. The variable *lnInd* represents industrial upgrading, measured as the ratio of tertiary to secondary industry output, following Shi and Wang (2024). Results in Table 6, column (2), show that HSR opening facilitates poverty reduction by driving local industrial transformation.

Table 6
Economic vitality mechanism analysis results.

	(1)	(2)
Dependent variable:	*lnnew_ent*	*lnInd*
HSR	0.120***	0.029***
	(9.91)	(3.32)
Control variables	YES	YES
Observations	9510	9510
R square	0.811	0.720
Number of households	1902	1902
Household fixed effects	YES	YES
Year fixed effects	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

In Table 7, the variable *lnEmpl* represents employment at the municipal level.[3] It is measured by the ratio of year-end employed population to the city's total population, reflecting the degree of employment attainment among the urban population. Column (1) in Table 7 shows that HSR opening can increase local employment rate. In addition, we also tested the impact of HSR on employment population in the primary, secondary, and tertiary sectors, which is represented by *lnEmp_primary*, *lnEmp_secondary*, and *lnEmp_tertiary*, respectively. Columns (2)–(4) show that HSR opening notably increases employment population in the secondary and tertiary sectors. It indicates that HSR facilitates labor transfer to non-agricultural sectors and promotes local industry development. HSR expands the labor market and job opportunities, enabling workers to improve income and access to social resources through interregional or intersectoral employment (Deng et al., 2019; Liang et al., 2020; Shao et al., 2017; Zhang and Xu, 2023, 2024).

We further tested the impact of HSR opening on labor transfer. Specifically, we calculate the number of adult household members engaged in migrant work and define the household labor transfer rate (*Hltr*) as proportion of household labor migration for employment (Huang et al., 2023; Wei and Wei, 2023). Column (5) in Table 7 shows that the impact of HSR opening has a positive impact on the household labor transfer rate, although the coefficient is insignificant. Still, it indicates the HSR can help individuals to choose to work outside their home cities. As the impact of HSR on labor transfer can vary among large, medium and small cities, we further investigated the HSR impact among these subgroups, which is in shown in Appendix Table A1.

5.4. Heterogeneity analyses

5.4.1. Household level

This segment of the study delves into the impact of HSR on household characteristics, particularly differentiating between urban and rural households, and further assessing the effects on households characterized by migration and the type of employment.

In Table 8,[4] the opening of HSR significantly alleviates the relative poverty of both urban and rural households (see columns (1) and (2)). The opening of HSR can help urban and rural residents to increase their mobility. Although rural residents may have less access to HSR compared with urban residents, HSR can also significantly reduce their relative poverty from their previous status.

In households with migrant members, as shown in columns (3) and (4) of Table 8, the impact of HSR on reducing relative poverty is more pronounced compared to households without migrant members. This finding underscores the importance of HSR in facilitating labor mobility, particularly for migrant workers who are often employed in cities far from their rural homes. HSR enables these workers to seek better employment opportunities, thereby enhancing their economic status and reducing the incidence of relative poverty.

Further analysis indicates a significant reduction in relative poverty among households with non-agricultural employment members (see columns (5) and (6)). It suggests that HSR particularly benefits those

[3] The regression results of the opening of HSR on the household employment level of residents' are near 10 % significant level, thus we only reported the employment rate at the municipal level.

[4] Household characteristics may change over the study period—for example, a shift from rural to urban registration, the emergence of migrant work, or a transition to non-agricultural employment. These changes may result in some households being included in more than once in subsample regression.

Table 7
Labor market mechanism analysis results.

	(1)	(2)	(3)	(4)	(5)
Dependent variable:	*lnEmpl*	*lnEmp_primary*	*lnEmp_secondary*	*lnEmp_tertiary*	*Hltr*
HSR	0.034***	−0.014	0.079***	0.016*	0.003
	(4.29)	(−1.58)	(6.58)	(1.79)	(0.31)
Control variables	YES	YES	YES	YES	YES
Observations	9510	9510	9510	9510	9510
R square	0.299	0.237	0.383	0.248	0.116
Number of households	1902	1902	1902	1902	1902
Household fixed effects	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

Table 8
Heterogeneity analysis results at the household level.

	(1)	(2)	(3)	(4)	(5)	(6)
Subsample:	Urban	Rural	With migrant worker	Without migrant worker	With non-agricultural employment	Without non-agricultural employment
Dependent variable:	*MRP*	*MRP*	*MRP*	*MRP*	*MRP*	*MRP*
HSR	−0.017**	−0.014*	−0.032***	−0.017**	−0.016**	−0.015
	(−1.97)	(−1.74)	(−2.96)	(−2.08)	(−2.41)	(−0.66)
Control variables	YES	YES	YES	YES	YES	YES
Observations	4013	5497	3998	5512	7956	1554
R square	0.123	0.099	0.106	0.121	0.115	0.157
Number of households	874	1188	1491	1775	1901	783
Household fixed effects	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

engaged in or seeking employment outside the agricultural sector. For these households, HSR not only provides access to a wider range of job opportunities but also supports more stable and potentially higher-income roles in industrial and service sectors. This effect is pivotal, as it highlights the role of HSR in supporting economic diversification and resilience, reducing dependence on local, often fluctuating agricultural incomes.

5.4.2. Municipal and regional levels

The impact of HSR on household multidimensional relative poverty varies significantly across different urban hierarchies and city sizes within China, reflecting disparities in economic development and social service provisions. We conducted the sub-sample analyses based on city hierarchy[5] and size.

Regarding city hierarchy, in the third, fourth, and fifth tier cities, the coefficient of *HSR* is negative (see column (2) in Table 9). However, in first and second tier cities, the coefficient in column (1) is not significant. This pattern suggests that HSR potentially plays a more crucial role in lower-ranked cities, which might have previously suffered from less connectivity and economic stagnation.

Concerning city size, in small cities, the opening of HSR has a significant effect on reducing the household multidimensional relative poverty (see column (5)), but not on large and medium cities (see column (3) and (4)). This observation aligns with prior research indicating that HSR tends to have more pronounced economic and social benefits in areas previously limited by poorer accessibility and fewer economic opportunities (Li et al., 2020; Wu et al., 2022; Zhang et al., 2023). This result is also consistent with the poverty reduction evidence from highway, which is also more significant in non-municipal counties (Tian et al., 2024).

Given China's vast territorial expanse and significant regional disparities in natural resource endowments and economic conditions, the samples were further divided into eastern, central, and western regions

Table 9
Heterogeneity analysis results at the city level.

	(1)	(2)	(3)	(4)	(5)
Subsample:	First and second tier cities	Third, fourth, and fifth tier cities	Large cities	Medium cities	Small cities
Dependent variable:	*MRP*	*MRP*	*MRP*	*MRP*	*MRP*
HSR	−0.017	−0.017***	−0.013	−0.006	−0.058***
	(−0.97)	(−2.63)	(−1.45)	(−0.63)	(−3.30)
Control variables	YES	YES	YES	YES	YES
Observations	1710	7800	3820	3420	2270
R square	0.110	0.109	0.093	0.129	0.155
Number of households	342	1560	764	684	454
Household fixed effects	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

[5] The first-tier cities include samples from first-tier and new first-tier cities including 19 cities including Beijing and Shanghai. The second-tier cities include samples from 30 cities including Kunming and Shenyang. The third-tier cities include samples from 70 cities including Taizhou and Luoyang. The fourth-tier cities include samples from 90 cities such as Yibin and Anshan, and the fifth-tier cities include samples from 128 cities such as Yanbian and Guigang. A total of 337 cities are involved, covering the sample of this study.

(Li et al., 2020). As poverty-stricken regions may have higher proportion of residents belong to relative poverty, we further conducted the heterogeneity analyses based on living in poverty-stricken regions or not.[6]

In Table 10, it is evident that the opening of HSR significantly reduces household multidimensional relative poverty in the central and western regions, with a notably stronger effect in the western regions (see columns (1) to (3)). These findings suggest that HSR significantly benefits less developed and peripheral areas, likely by enhancing connectivity, which facilitates economic integration and access to broader markets and services. However, the effect of HSR in the eastern region is insignificant, similar to the research on the impact of highways on poverty (Tian et al., 2024). Moreover, results in columns (4)–(5) confirm that the HSR opening is slightly more conducive to reducing household multidimensional relative poverty in poverty-stricken areas, implicating that HSR contributes to consolidating absolute poverty alleviation achievements.

5.5. Robustness checks

5.5.1. Stacked DID method

When the policy implementation time is highly dispersed across different individuals, the multi-period DID may suffer from bias due to heterogeneous treatment effects. The Stacked DID method is used to evaluate policies implemented in batches, addressing issues arising from multiple treatment periods (Baker et al., 2022). Stacked DID constructs multiple "pseudo-experiment periods" by stacking each treatment group with its specific control group subsample into a large panel, and then performing DID estimation to obtain the overall treatment effect. This method helps resolve the bias that may arise from the two-way fixed effects model in multi-period DID when treatment effects are heterogeneous, thereby improving the robustness of the estimates.

To mitigate potential biases associated with the multi-period DID approach, this study employed a stacked DID method as an additional robustness check (Baker et al., 2022). In columns (1)–(2) in Table 11, the regression results align with the baseline results and parallel trend test results, verifying the robustness of previous results.

Table 10
Heterogeneity analysis results at the regional level.

	(1)	(2)	(3)	(4)	(5)
Subsample:	Eastern region	Central region	Western region	Poverty-stricken areas	Non poverty-stricken areas
Dependent variable:	*MRP*	*MRP*	*MRP*	*MRP*	*MRP*
HSR	0.008 (0.93)	−0.030*** (−3.14)	−0.031* (−1.89)	−0.017** (−1.97)	−0.014* (−1.65)
Control variables	YES	YES	YES	YES	YES
Observations	3705	3330	2475	5175	4335
R square	0.090	0.137	0.124	0.131	0.091
Number of households	741	666	495	1035	867
Household fixed effects	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

Table 11
Stacked DID regression results.

	(1)	(2)
Dependent variable:	*MRP*	*MRP*
HSR	−0.019*** (−2.94)	
The fourth year before the opening of HSR (*pre2*)		0.006 (0.52)
The second year before the opening of HSR (*pre1*)		−0.008 (−0.70)
The year of the opening of HSR (*current*)		−0.022* (−1.89)
The second year after the opening of HSR (*post1*)		−0.017 (−1.33)
The fourth year after the opening of HSR (*post2*)		−0.031** (−2.31)
The sixth year and beyond after the opening of HSR (*post3+*)		−0.029** (−1.99)
Control variables	YES	YES
Observations	19,310	19,310
R-squared	0.608	0.608
Household FE	YES	YES
Year FE	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

5.5.2. Instrumental variable (IV) analysis

Addressing potential endogeneity between HSR development and household multidimensional relative poverty, we utilized instrumental variables that are geographically exogenous yet theoretically relevant to the construction and operation of HSR. Infrastructure location selection by governments may be related to the local economic and political priorities (Lu et al., 2022b; Ma, 2022). Following previous research, we overcome this by using instrumental variables to reduce these economic or political impact (Lu et al., 2023). Specifically, we adopt two natural geographic indicators—terrain ruggedness index and river density—as instrumental variables to mitigate potential economic and political endogeneity in HSR investment decisions (Chen and Chen, 2023; Liu and Lin, 2025; Nunn and Puga, 2012; Shen et al., 2023; Wu et al., 2023; Yan et al., 2023).

First, HSR construction requires high requirement for terrain conditions (Chen and Chen, 2023; Yan et al., 2023). For instance, the maximum gradient for HSR construction is below 1.5–2 % to maintain speed and energy efficiency. The terrain ruggedness index, derived from elevation variations, captures the degree of surface undulation and complexity, indicates the possibility to build HSR (Nunn and Puga, 2012; Shen et al., 2023; Wu et al., 2023; Yan et al., 2023). The place with higher terrain ruggedness index is more difficult to meet the HSR construction conditions.

Second, river density can influence HSR in two ways. One the one hand, the need for HSR in a place with higher river density can be lower as the waterway can solve some transport demand already. One the other hand, river density can increase the HSR construction difficulty, due to unstable terrain and flood risks, increasing costs and rerouting HSR projects (Chen and Chen, 2023; Liu and Lin, 2025; Qie et al., 2022). River density is defined as the total river length per unit area, it reflects the complexity of the hydrological system. As the instruments are time-invariant, we used their interaction with time trend variable for analysis (Hanley et al., 2022).

In Table 12, the first-stage regression results of the IV regression indicate a significant correlation between the selected instrumental variables and the core independent variable (*HSR*). The second-stage regression results of the IV regression show that the coefficient of *HSR* is significantly negative and is larger than the baseline regression coefficient of −0.018. This indicates that the IV regression results are

[6] According to the list of 832 poverty-stricken counties determined by the Chinese government (including 832 key counties for national poverty alleviation and development, as well as counties in concentrated and contiguous areas of extreme poverty), we divide cities into poverty-stricken areas and non poverty-stricken areas.

Table 12
IV regression results.

	(1)	(2)
	First stage	Second stage
Dependent variable:	*HSR*	*MRP*
HSR		−0.107**
		(0.052)
*Terrain ruggedness index*trend*	−2.094***	
	(0.365)	
*River density*trend*	−0.277***	
	(0.063)	
Control variables	YES	YES
Observations	9510	9510
R square	–	−0.023
Household fixed effects	YES	YES
Year fixed effects	YES	YES
F-statistic value	24.96	–
The 10 % critical value of the Stock-Yogo weak ID test	19.93	
Over-identification test	–	0.3946
Endogeneity test	–	0.0219

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

consistent with the baseline results, suggesting the baseline regression results are robust. A problem in IV regression is that the instrument is only weakly correlated with the endogenous explanatory variable. This can lead to biased IV estimates and unreliable inference. To detect weak instruments, the standard approach is to check the first-stage F-statistic. In the weak instrumental variable test (KP-F test), the F-statistic value is 24.96 greater than 10 and the critical value of 10 % (19.93), rejecting the weak instrument hypothesis. Moreover, the result also passed the over-identification test (Hansen J > 0.1) and endogeneity test.

5.5.3. Placebo test

Additionally, we conducted the placebo test by randomizing both the policy implementation time and the treatment group assignments. Based on the double random placebo test method, we generated pseudo policy and treatment group dummy variables. The rationale behind the placebo test is to ensure that the significant relationships identified in the actual data are not merely artifacts of the model or due to random chance.

As shown in Fig. 4, the vertical red dashed line represents the coefficient of the HSR dummy variable from the baseline model (−0.018), while the horizontal red dashed line marks the *p*-value threshold of 0.1. The blue dots depict the *p*-values obtained from the placebo tests, and the red curve illustrates their density distribution. Two key criteria support the robustness of the regression results. First, the pseudo-regression coefficients are approximately normally distributed with a mean close to zero, indicating no significant effects under the placebo assumption. Second, the majority of the corresponding p-values exceed the 0.1 threshold, suggesting that the placebo estimates are generally insignificant across different hypothetical treatment periods. In addition, the baseline coefficient of −0.018 lies in the left tail of the placebo distribution and falls outside the range of most pseudo-regression estimates, identifying it as an outlier. These results collectively confirm the robustness of the empirical findings illustrated in Fig. 4.

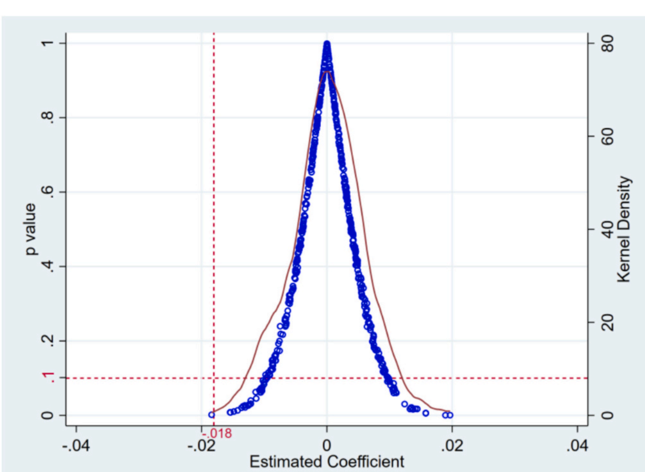

Fig. 4. Placebo test results.

5.5.4. Other robustness tests

The additional robustness tests include: (1) Propensity Score Matching-DID (PSM-DID) method. The DID method assumes comparability between the treatment and control groups. However, differences in initial conditions may lead to biased estimates. PSM-DID was applied to address potential sample selection bias, using a 1:2 caliper nearest neighbor matching approach across five survey periods. This method first estimates the propensity scores of being in the treatment group using a Logit model. Then, each treated unit is matched with a control unit with the closest propensity score. Finally, the balance of covariates between the matched treatment and control groups is assessed to ensure comparability. The balance tests and kernel density plots of propensity scores, shown in Table A2 and Fig. A1 respectively, confirm the adequacy of the matching process. (2) Excluding developed cities. Referring to the approach of Yu et al. (2020), the analysis was repeated after excluding samples from provincial capitals, municipalities directly under the central government, sub-provincial cities, and the three major economic zones, ensuring that the results are not disproportionately influenced by the most economically developed areas. (3) Including the HSR station number. Recognizing that cities with multiple HSR stations might offer greater accessibility and consequently different economic impacts, the cumulative number of HSR stations servicing “G” trains was added as a control variable. (4) Setting weights using the entropy method. To ensure the robustness of key assumptions based on the A-F method, particularly with respect to the assignment of weights and the selection of indicators, we performed a sensitivity analysis employing the entropy weighting method. (5) Considering clustering robust standard errors. To further address the issue of serial correlation, we incorporate clustered standard errors into the baseline regression. (6) Considering city fixed effects. By controlling for city fixed effects in addition to household and year fixed effects, the regression further absorbs all time-invariant city characteristics, thereby enhancing the reliability of policy effect identification. In columns (1)–(6) in Table 13, the robustness test results are consistent with the baseline regression results, underscoring the robustness of the original conclusions.

6. Discussion

This study utilizes the CFPS database to investigate the influence of HSR on household multidimensional relative poverty, contributing significantly to the existing body of knowledge in several ways. First, this research enriches the empirical landscape by shifting focus from absolute to relative poverty, broadening the understanding beyond mere economic dimensions to include health, lifestyle, and subjective well-being (Li et al., 2020; Wu et al., 2022; Zhang et al., 2023). This approach not only offers a more holistic view of poverty but also aligns with contemporary movements toward understanding poverty in a multidimensional space, thereby enhancing the relevance and applicability of poverty metrics.

Second, this study reveals that the opening of HSR has a pronounced effect on reducing relative poverty, particularly by fostering new business development, driving industrial upgrading, enhancing local employment opportunities, and facilitating labor mobility. This finding aligns with previous research on local economic development (Lan et al., 2023; Liang et al., 2020; Liu and Lin, 2025; Yi et al., 2025). The result concerning employment also echoes discussions in the labor market literature (Di Matteo and Cardinale, 2023; Dong, 2018; Heuermann and

Table 13
Other robustness check results.

	(1) PSM-DID method	(2) Excluding developed cities	(3) Including HSR station number	(4) Setting weights using entropy method	(5) Considering clustering robust standard errors	(6) Considering city fixed effects
Dependent variable:	*MRP*	*MRP*	*MRP*	*MRP_new*	*MRP*	*MRP*
HSR	−0.016**	−0.024***	−0.015**	−0.014***	−0.018***	−0.018***
	(−2.47)	(−3.28)	(−2.35)	(−2.77)	(−3.00)	(−3.11)
HSR_station	–	–	−0.001	–	–	–
	–	–	(−0.89)	–	–	–
Control variables	YES	YES	YES	YES	YES	YES
Observations	7977	6700	9510	9510	9510	9510
R square	0.112	0.117	0.107	0.102	0.107	0.614
Number of households	–	1340	1902	1902	1902	1902
City fixed effects	NO	NO	NO	NO	NO	YES
Household fixed effects	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

Schmieder, 2019; Kaewunruen et al., 2019; Zhang and Xu, 2024). These findings suggest that HSR can serve as a critical infrastructure component that enables residents to access broader employment markets, both locally and in larger urban centers.

Third, another significant finding is that HSR particularly aids low-tier, small cities, and regions traditionally viewed as economically disadvantaged, such as the central and western parts of China and designated poverty-stricken areas. The reason can be residents in these cities or areas with disadvantages in size or locations can mobile to large cities to improve their income, which also benefits their economic, life, and subjective aspects significantly. This suggests that HSR can be an equalizing force, mitigating locational and size-related disadvantages and offering substantial economic and social benefits to residents in these areas.

7. Conclusion

7.1. Key findings

The main findings of this study are as follows: Firstly, this study concludes that the introduction of HSR significantly curtails household multidimensional relative poverty, with effects enduring over time. Specifically, the relative poverty levels in cities with HSR access have decreased by an average of 1.8 %, highlighting the substantial impact of HSR opening. Secondly, the opening of HSR effectively reduces relative poverty by stimulating local economic vitality and expanding employment opportunities. Third, the impact of HSR opening on relative poverty varies across households, cities, and regions. Specifically, the opening of HSR is especially conducive to reducing the household multidimensional relative poverty in low rank cities, small cities, central and western regions, as well as poverty-stricken areas. Lastly, households with migrant workers or households with non-agricultural employment can benefit more from the opening of HSR, and the relative poverty status of rural households can also be improved.

7.2. Policy implications

Firstly, HSR can serve as a potential tool for alleviating household multidimensional relative poverty, especially for small cities, low rank cities, central and western regions, as well as poverty-stricken areas. Thus, policymakers can consider promoting a more balanced spatial distribution of HSR stations in future plans. Secondly, as the HSR plays a significant role in reducing the relative poverty of households with migrant workers or households with non-agricultural employment, the government needs to improve the connectivity between HSR stations and these groups. For instance, the rural transportation facilities can be improved to increase the accessibility of these residents to HSR stations. Thirdly, the HSR opening reduces households' relative poverty by increasing labor transfer and local employment. It is necessary for policy makers to facilitate information dissemination about job vacancies and economic opportunities created by these infrastructural enhancements.

7.3. Research limitations and outlook

This study has several limitations. First, there may be additional mechanisms through which HSR influences household multidimensional relative poverty. However, data limitations in the CFPS database restrict a more in-depth investigation of these mechanisms, particularly regarding the reasons behind the differentiated impacts of HSR opening on labor mobility across different sectors. Second, this study may be limited by the time frame covered by the CFPS data. For future research, as the government has paid more attention to rural road construction in recent years, suggesting that further research could extend to the impacts of rural road developments on relative poverty alleviation. Additionally, this research focuses on household level relative poverty, future research could examine the effects of transportation infrastructure on individual-level poverty, providing another view of the social benefits of such investments. In particular, future access to more granular household- or individual-level data, including sectoral employment details, sources of job creation, and initial labor status, would allow for a more precise analysis of how transportation infrastructure improvements mitigate relative poverty.

CRediT authorship contribution statement

Yaning Hu: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Haiyan Lu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Hao Chai:** Validation, Software, Formal analysis. **Xinyang Gao:** Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix

A.1. Sub-sample test of labor transfer mechanism

We further investigated the impact of HSR on labor transfer in cities of different sizes. According to the classification standard based on the urban population, with cities having a permanent population of over 1 million classified as large cities, cities with a population of 0.5 million to 1 million as medium-sized cities, and cities with a population below 0.5 million as small cities. The large city sample includes super-large cities, mega cities, type I and type II large cities in China. The medium-sized city sample includes 135 medium-sized cities such as Jiaxing, Huzhou, and Jinhua. The small city sample includes the remaining cities.

The subsample regression results are shown in Table A1. In medium and small cities (see column (3)), the coefficients of HSR are significantly positive at the 1 % level, suggesting that HSR facilitates the mobility of labor toward regions offering more or better job opportunities. In contrast, large cities, with their more robust job markets, benefit differently from HSR. In column (2), the coefficient of HSR is significantly negative at the 5 % level. This indicates that HSR in larger cities tends to concentrate more workers by enhancing accessibility and attractiveness for both labor and firms. These analyses show that the labor transfer can vary in various cities. This results is consistent with the discussion on agglomeration effects brought by HSR in large cities (Jiao et al., 2025), and the diffusion effect in medium and small cities.

Table A1 Sub-sample test results of labor transfer mechanism.

	(1)	(2)	(3)
Subsample:	Full sample	Large cities	Medium and small cities
Dependent variable:	*Hltr*	*Hltr*	*Hltr*
HSR	0.003	−0.030**	0.033***
	(0.31)	(−2.38)	(2.91)
Control variables	YES	YES	YES
Observations	9510	3820	5690
R square	0.116	0.105	0.139
Number of households	1902	764	1138
Household fixed effects	YES	YES	YES
Year fixed effects	YES	YES	YES

(1) ***, **, and * indicate passing the 1 %, 5 %, and 10 % significance test respectively; (2) Robust standard error values are in parentheses.

A.2. Test process of PSM-DID method

Table A2 Balance test results.

	Before matching					After matching				
	2011b	2013b	2015b	2017b	2019b	2011a	2013a	2015a	2017a	2019a
lnSize	−0.198	−0.145	−0.107	0.161	−0.022	−0.273	−0.114	−0.050	0.064	−0.175
	(−1.23)	(−0.96)	(−0.72)	(1.11)	(−0.16)	(−1.41)	(−0.61)	(−0.27)	(0.38)	(−1.07)
lnAge	−0.166	−0.248	−0.262	−0.348	−0.589**	−0.426	−0.034	−0.431	0.040	−0.062
	(−0.55)	(−0.79)	(−0.88)	(−1.16)	(−1.96)	(−1.13)	(−0.09)	(−1.19)	(0.11)	(−0.18)
Fhcl	−0.016	−0.007	−0.227	−0.236	−0.260*	0.061	0.164	0.063	0.076	0.029
	(−0.10)	(−0.05)	(−1.50)	(−1.53)	(−1.83)	(0.30)	(0.81)	(0.34)	(0.41)	(0.17)
lnFin	0.024	0.028**	−0.016	−0.012	−0.017	0.007	0.017	0.000	0.021	0.009
	(0.99)	(2.43)	(−1.24)	(−0.89)	(−1.28)	(0.24)	(1.29)	(0.03)	(1.30)	(0.58)
lnSoc	0.082	0.010	0.142**	0.095*	0.136***	−0.028	−0.007	0.021	0.056	0.051
	(1.48)	(0.20)	(2.14)	(1.75)	(2.59)	(−0.45)	(−0.11)	(0.29)	(0.89)	(0.88)
lnLand	−0.039**	−0.040***	−0.065***	−0.068***	−0.050***	−0.016	0.011	−0.011	−0.003	−0.003
	(−2.50)	(−2.84)	(−4.71)	(−4.87)	(−3.74)	(−0.89)	(0.69)	(−0.71)	(−0.20)	(−0.19)
lnTech	0.610***	0.450***	0.117	0.086	0.440***	−0.064	−0.062	−0.006	−0.140	0.042
	(5.37)	(3.79)	(1.07)	(0.91)	(5.65)	(−0.46)	(−0.48)	(−0.05)	(−1.25)	(0.46)
lnFgn	17.987***	17.165***	29.248***	32.727***	39.865***	3.583	6.188**	5.421	11.058***	7.601
	(5.84)	(5.64)	(7.18)	(8.51)	(8.53)	(1.22)	(2.18)	(1.22)	(2.79)	(1.41)
lnExp	1.208***	1.249***	1.314***	1.327***	0.784***	0.364*	0.358*	0.313*	0.507***	0.338*
	(7.43)	(7.47)	(8.25)	(7.45)	(4.68)	(1.91)	(1.84)	(1.75)	(2.71)	(1.87)
N	1902	1902	1902	1902	1902	839	883	845	866	885

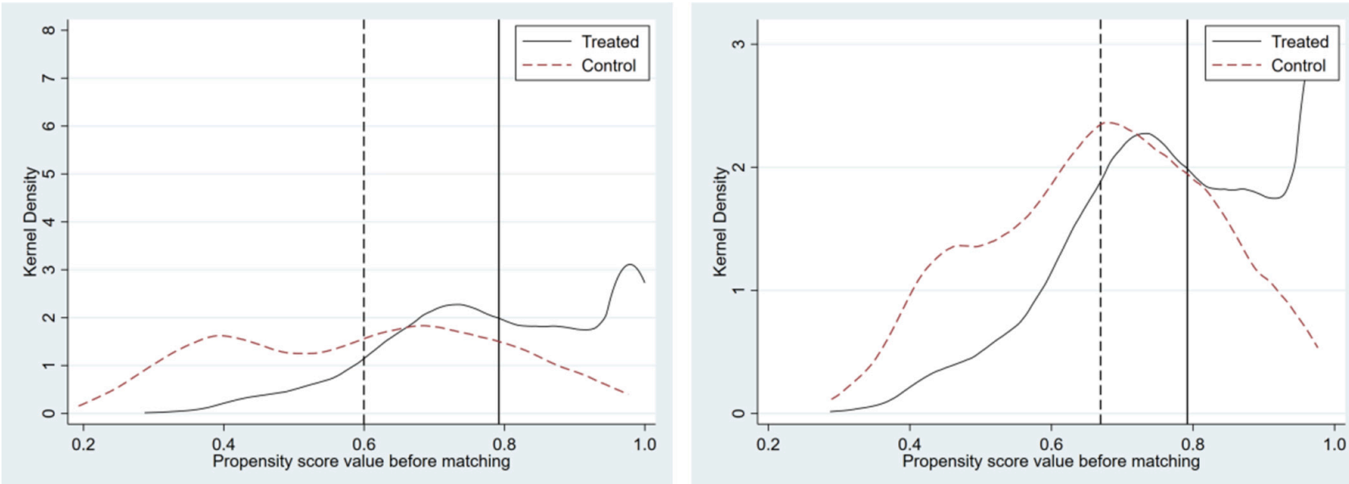

Fig. A1 Comparison of kernel density distribution before and after PSM.

Data availability

The data that has been used is confidential.

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